

Data Collection and Preprocessing Phase

Date	20 September 2026
Team ID	
Project Title	Food Consumer Behavior Analytics Portal
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	Dataset contains 50,000 food transaction records used to analyze consumer purchasing patterns across regions, age groups, meal types, spending patterns, and feedback ratings. The data is used for the Tableau dashboard.
Data Cleaning	Checked the dataset for missing values, duplicate records, inconsistent entries, and schema issues. Records/fields requiring cleaning were standardized before visualization so that the dashboard uses reliable data.
Data Transformation	Created the Age Group calculated field using IF/ELSEIF logic to categorize customer ages. Prepared fields required for meal-type revenue, city-wise order distribution, demographic analysis, regional demand, order frequency, and spending analysis.
Data Type Conversion	Verified and corrected field data types so categorical fields are treated as dimensions and numeric/measure fields are available for aggregation and visualization. Date fields, where present, were handled as date values.
Column Splitting and Merging	No major column splitting or merging was required for the final dashboard. Existing fields were used directly, while calculated/derived fields such as Age Group were created where required.

Data Modeling	The Tableau workbook uses the prepared transaction dataset as the analytical source. Calculated fields and aggregations support the nine dashboard sheets, including revenue by meal type, city-wise order distribution, age-group analysis, regional demand, order frequency, and spending patterns.
Save Processed Data	The cleaned/prepared dataset was used as the Tableau data source and the completed workbook/dashboard was published to Tableau Public. The processed analytical output was then integrated into the Flask web application.